Dual-Domain Adversarial Purification for Robust Remote Sensing Scene Classification

Yuru Su, Shaohui Mei, *Senior Member, IEEE*, Mingyang Ma, Shuai Wan, *Member, IEEE*, and Jiao Shi

***Abstract*—Deep learning has boosted remote sensing (RS) scene classification, but adversarial examples can still cause high-confidence misclassification with imperceptible perturbations. Adversarial purification (AP) offers a practical test-time defense without retraining the classifier. However, most existing methods are confined to pixel-space restoration, which may leave residual adversarial effects that persist and amplify through feature extraction, ultimately biasing the prediction. To address these issues, a dual-domain AP (DDAP) framework is proposed to mitigate adversarial effects at both the pixel and feature levels in a unified pipeline. In the pixel domain, a pixel-domain frequency-aware diffusion purification (PFDP) module performs diffusion-based restoration through a frequency-aware dual-stream U-Net (FD-UNet). By integrating adaptive spectral filtering with multidomain consistency constraints, PFDP reduces adversarial-perturbation-dominated high-frequency responses while preserving structural details and semantic information in RS imagery. In the feature domain, an adversarial vulnerable channel dropout (AVCD) strategy models unshifted shallow-feature statistics with a Gaussian mixture model (GMM) and adaptively assigns channelwise dropout probabilities based on a samplewise shift score and channel vulnerability, thereby suppressing residual adversarial influence before downstream classification. Extensive experiments on UC Merced (UCM) and aerial image dataset (AID) across multiple backbones and attack types demonstrate that DDAP consistently improves robustness while maintaining a favorable clean–robust balance compared with representative baselines.**

***Index Terms*—Adversarial purification (AP), adversarial robustness, diffusion models, Gaussian mixture model (GMM).**

I. Introduction

IN RECENT years, advances in artificial intelligence (AI) techniques have continuously strengthened intelligent interpretation in Earth observation, where deep learning has become a key approach across a broad range of remote sensing (RS) applications [1], [2], driving sustained progress in land cover classification [3], [4], disaster interpretation [5], [6], and urban management [7], [8]. However, these performance gains have also raised growing concerns about security and robustness, as deep neural networks remain highly vulnerable to adversarial examples. By adding imperceptible perturbations, adversarial examples can be crafted to induce the state-of-the-art networks to produce incorrect predictions with high confidence. More importantly, this vulnerability is not confined to a single data type, as adversarial examples have been empirically observed across multiple RS modalities, including optical imagery [9], synthetic aperture radar (SAR) imagery [10], and hyperspectral imagery [11], affecting a wide spectrum of tasks, such as scene classification [12], change detection [13], and object detection [2]. Such security risks substantially hinder the reliable deployment of intelligent RS systems in real-world complex scenarios.

To mitigate the security threats posed by adversarial examples, various defense strategies have been developed to defend against adversarial attacks and improve model robustness. The existing studies mainly follow two technical directions: adversarial training (AT) and adversarial purification (AP). AT [14] enhances robustness by incorporating adversarial samples into the training process and is widely regarded as an effective defense strategy. However, it often incurs substantial computational overhead and is typically tailored to specific attack types, which limits its generalization to unseen attacks. Meanwhile, the AP methods [15] have attracted increasing attention because they purify perturbed inputs during inference without retraining the classifier, offering greater flexibility and practicality for real-world deployment. For instance, Yang et al. [16] proposed AP with the manifold hypothesis (APMH), a manifold-hypothesis-based purification framework that integrates manifold learning with variational inference to achieve adversarial robustness without expensive AT. Lin et al. [17] introduced AT on purification (AToP), which combines AT with AP to better balance robustness against known attacks, generalization to unseen attacks, and standard accuracy on clean samples. Yu et al. [18] presented a universal adversarial defense method for RS (UAD-RS) imagery based on a pretrained denoising diffusion model, where Gaussian noise is injected to disrupt adversarial perturbations and the reverse denoising process is exploited to purify the inputs. Lei et al. [19] further proposed one-step control purification (OSCP), enabling robust diffusion-based AP with only a single neural function evaluation (NFE).

The existing AP methods, including recent diffusion-based approaches, generally aim to restore adversarial inputs toward the clean data distribution before classification. These methods mainly perform input-level correction through denoising, reconstruction, or guided generation to reduce the influence

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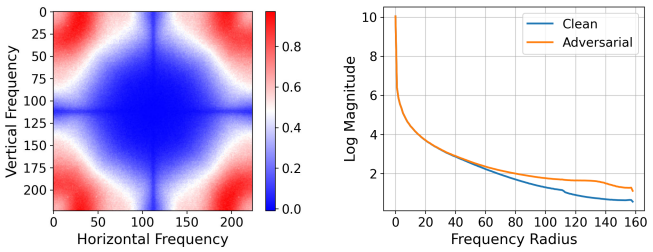


Fig. 1. Frequency-domain observation of adversarial perturbations. (a) Averaged spectrum difference between adversarial and clean samples. (b) Radially averaged log spectrum profiles.

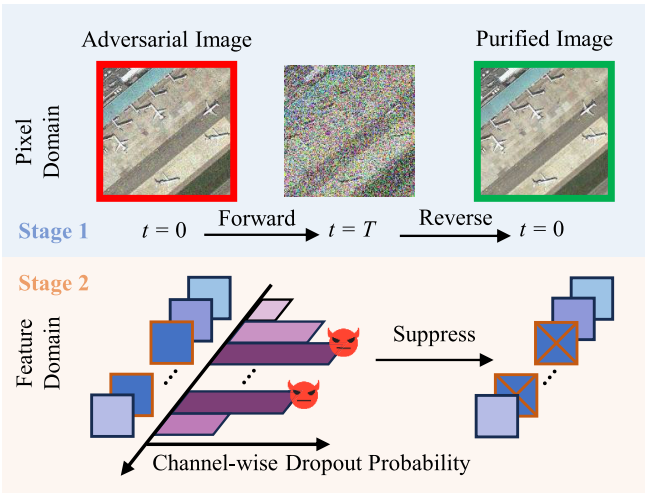


Fig. 2. Illustration of the proposed dual-domain defense pipeline, consisting of pixel-domain image purification and feature-domain suppression of adversarial effects.

of adversarial perturbations. Although they have achieved promising robustness, most of them are still designed around pixel-space restoration. However, pixel-domain purification alone often cannot fully eliminate adversarial effects, and a small amount of perturbation may still remain after input restoration. During inference, these residual components can accumulate and be progressively amplified through successive feature extraction layers, leading to channelwise abnormal activations that ultimately bias the classifier toward incorrect predictions. Meanwhile, in the RS setting, as illustrated in Fig. 1, adversarial samples exhibit stronger spectral responses than clean samples in the middle-to-high frequency region. This observation suggests that adversarial perturbations are distributed nonuniformly in the pixel domain, motivating a purification strategy that places greater emphasis on the middle-to-high-frequency bands.

Based on the above analysis, the effective adversarial defense should not be confined to a single domain. Instead, it should jointly consider input restoration and feature-level robustness within a unified pipeline. To this end, a dual-domain AP (DDAP) framework is proposed to systematically mitigate adversarial effects at both the pixel and feature levels, as shown in Fig. 2. In the pixel domain, a pixel-domain frequency-aware diffusion purification (PFDP) module is introduced, where the reverse diffusion denoiser is implemented with a frequency-aware dual-stream U-Net (FD-UNet). FD-UNet uses adaptive frequency-domain filtering to extract perturbation-salient high-frequency responses and route them to the decoder via skip connections as explicit auxiliary cues for identifying and predicting adversarial components, while the main pathway focuses on estimating the residual stochastic noise for stable denoising. Moreover, multidimensional consistency constraints further regularize the reverse diffusion process, helping preserve structural details and semantic content in RS imagery. In the feature domain, an adversarial vulnerable channel dropout (AVCD) strategy is further developed. It models the distribution of unshifted features using a Gaussian mixture model (GMM) to quantify the feature shift of the input sample and estimates channelwise vulnerability by measuring perturbation-induced response variations. Using both the sample-level shift score and channel-level vulnerability, AVCD adaptively assigns channelwise dropout probabilities to selectively suppress high-risk channels. This design limits the propagation of residual adversarial effects after pixel-domain diffusion purification, thereby establishing a robust dual-level defense.

Comprehensive experiments are conducted on two widely adopted benchmarks, UC Merced (UCM) land-use [20] and the aerial image dataset (AID) [21], under five representative attacks, including FGSM [22], I-FGSM [23], PGD [14], CW [24], and Jitter [25]. The results demonstrate that the proposed DDAP framework achieves consistently strong performance across three representative backbone networks, namely, MobileNet-V3, ResNet-50, and DenseNet-121. The main contributions are summarized as follows.

1) A DDAP framework is proposed to overcome the limitations of single-domain defenses by jointly performing purification at both the pixel level and the feature level, thereby mitigating adversarial effects in a unified pipeline.
2) A PFDP module is developed by implementing the reverse diffusion process with an FD-UNet. This design enables perturbation-related high-frequency components to be explicitly modeled and predicted, while multidimensional consistency constraints help preserve the geometric structures and semantic content of RS imagery.
3) A feature-domain AVCD strategy is proposed to adaptively assign channelwise dropout rates based on the sample-level feature shift score and channelwise vulnerability. This mechanism reduces the propagation of residual adversarial effects after pixel-domain purification.

The rest of this article is organized as follows. Section II briefly reviews the related work. Section III details the proposed DDAP framework. Section IV describes the experimental setup and reports the comprehensive results. Conclusions are summarized in Section V.

II. Related Work

A. Adversarial Training

AT is a representative approach to improve adversarial robustness. It augments the training process with adversarial

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samples and jointly trains on them alongside clean data [14]. Although AT can substantially enhance robustness against known attacks, it also faces two practical challenges. First, robustness gains are often accompanied by reduced clean accuracy, reflecting an inherent tradeoff between standard performance and adversarial robustness. To alleviate this issue, TRADES [26] decomposed robust error into natural error and boundary error and derived a differentiable upper bound based on a classification-calibrated loss. More recently, Wang and Liu [27] showed that the accuracy–robustness tradeoff in AT primarily arises from overemphasizing hard adversarial examples rather than from insufficient learning. They further proposed robust perception AT (RPAT) to achieve a better balance between accuracy and robustness. Second, AT is sensitive to attack types, perturbation budgets, and hyperparameter settings used during training, which can lead to overfitting to specific attacks and poor generalization to unseen attacks. To improve generalization and training stability, recent studies have explored strategies, such as randomized interlayer processing [28], data augmentation [29], and distillation-based techniques [30]. Nevertheless, AT still incurs substantial training and tuning costs in practice.

B. Adversarial Purification

AP has received growing attention as a test-time defense that does not require retraining the classifier, providing a more deployment-friendly paradigm [15]. Early AP studies primarily relied on the generative power of GANs for input reconstruction. For example, Defense-GAN [31] learned the distribution of clean images and generated outputs that preserve semantic content while suppressing adversarial perturbations. Subsequently, the neural representation purifier (NRP) [32] adopted automatically derived self-supervision to remove adversarial corruption, whereas the adversarial-noise removing network (ARN) [33] emphasized learning attack-invariant features to suppress perturbations while retaining discriminative semantics. Beyond GAN-based methods, structural priors have also been incorporated. Yang et al. [16] built on the manifold hypothesis and integrate manifold learning with variational inference to improve robustness without expensive AT, resulting in APMH method. In parallel, Lin et al. [17] proposed AToP, which co-designs AT and purification to better balance robustness to known attacks, generalization to unseen attacks, and clean accuracy.

With the success of diffusion models in generative modeling, AP has further shifted toward diffusion-based purification. The representative methods, such as DiffPure [15], exploited the reverse denoising process to jointly mitigate adversarial perturbations and the Gaussian noise introduced in the forward diffusion trajectory. In the RS domain, Yu et al. [18] proposed UAD-RS based on a pretrained denoising diffusion model, where Gaussian noise is injected to disrupt adversarial structures and reverse denoising is used for universal purification. To address the high inference cost of multistep diffusion sampling, Lei et al. [19] introduced OSCP, which compresses purification into a single NFE for more efficient deployment. Owing to their strong denoising capability and stable training dynamics, diffusion models have become a mainstream choice for modern AP.

Unlike the existing purification methods, the proposed DDAP is not limited to generic pixel-space restoration. Instead, it explicitly considers both the spectral characteristics of adversarial perturbations in the pixel domain and the residual adversarial effects in the feature domain, thereby providing a more effective dual-domain defense mechanism.

C. Diffusion Models

Diffusion models are a class of probabilistic generative approaches that have recently achieved high sample quality and diversity in image synthesis [34]. Among them, denoising diffusion probabilistic models (DDPMs) [35] formulated generation as a gradual corruption and recovery procedure. In the forward phase, a Markovian transition progressively perturbs clean data into approximately Gaussian noise; in the reverse phase, the corresponding inverse Markov chain is learned to iteratively denoise and recover the data distribution. To accelerate sampling, the denoising diffusion implicit models (DDIMs) [36] reduced the number of denoising steps while maintaining generation fidelity. Latent diffusion models (LDMs) [37] further improved efficiency by conducting diffusion in the latent space of a pretrained autoencoder, substantially lowering training costs and facilitating high-resolution generation. Formally, given a clean sample $\mathbf{x}_0$, the forward diffusion process with T steps is defined as follows:

$$q(\mathbf{x}_1, \ldots, \mathbf{x}_T \mid \mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \tag{1}$$

which progressively corrupts the data by adding Gaussian noise at each step. The generative process that maps the Gaussian prior back to the data distribution is referred to as the reverse diffusion process. Given a latent variable $\mathbf{x}_t$, the reverse process also consists of T steps and can be written as follows:

$$p_\theta(\mathbf{x}_0, \ldots, \mathbf{x}_{T-1} \mid \mathbf{x}_T) = \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \tag{2}$$

where θ denotes the learnable parameters of the reverse transitions. In the standard DDPM, the parameterization is commonly implemented using an end-to-end U-Net, which models stepwise denoising transformations to progressively reconstruct $\mathbf{x}_0$.

III. Proposed Method

A. Overview

To defend against adversarial attacks in RS scene classification, an integrated framework termed DDAP is proposed, as illustrated in Fig. 3. DDAP mitigates adversarial impact on a unified pipeline by combining pixel-domain restoration with feature-domain refinement. Given an adversarial input, DDAP first performs pixel-domain restoration to reduce input-level corruption while preserving the geometric structures and semantic content of RS imagery. Since residual perturbation effects may persist after pixel-level processing and

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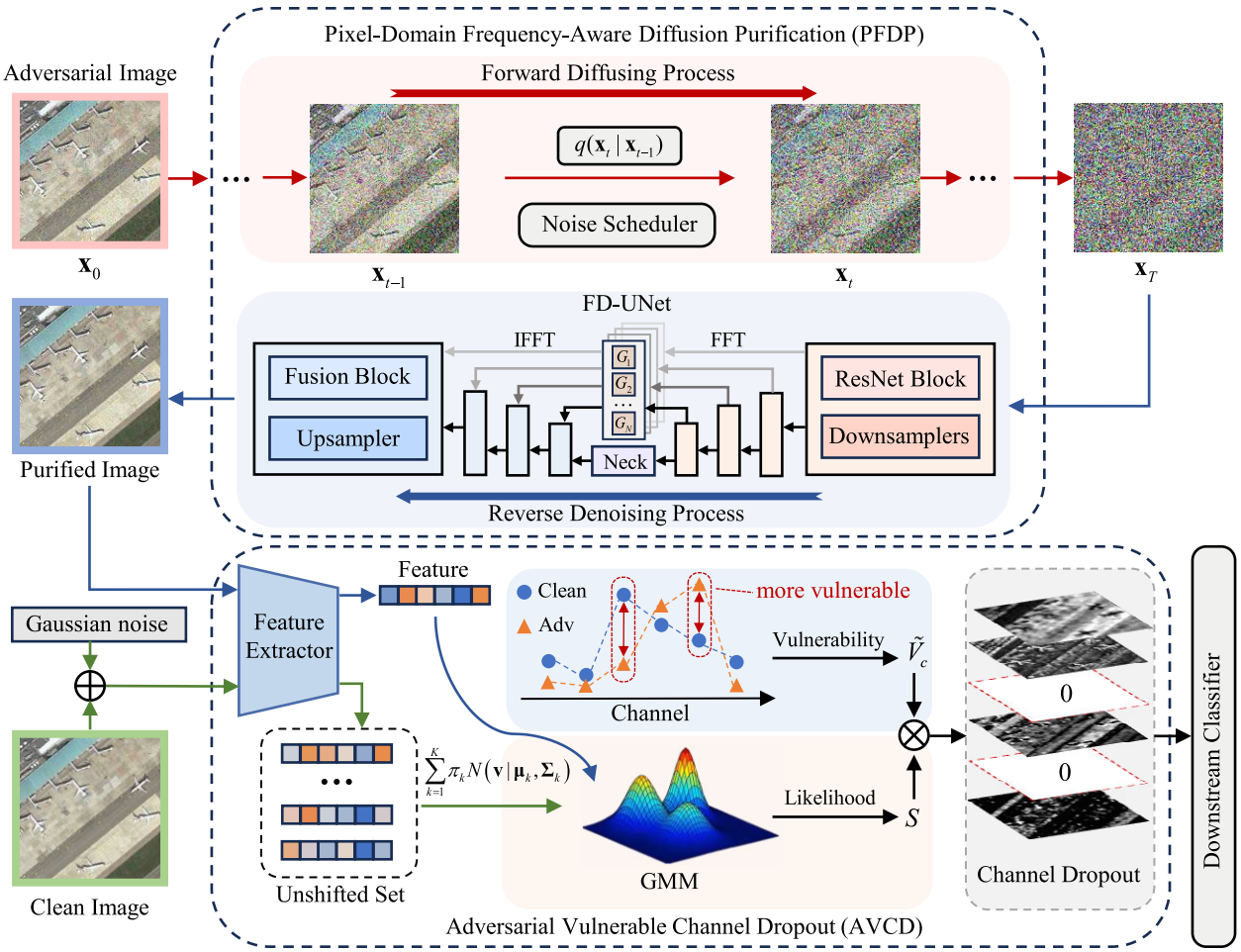


Fig. 3. Overview of the proposed DDAP framework. In the pixel domain, PFDP performs diffusion-based purification, where the forward process $q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$ gradually perturbs an adversarial input $\mathbf{x}_0$ into Gaussian noise $\mathbf{x}_T$, and the reverse denoising process reconstructs a purified image using FD-UNet. In the feature domain, AVCD extracts shallow features and computes a samplewise shift score S from the GMM log-likelihood together with a channel vulnerability estimate $\tilde{V}_c$, which jointly determine channelwise dropout rates before the downstream classifier.

can be amplified during feature extraction, DDAP further refines intermediate representations by suppressing channel-wise abnormal responses, preventing such residual effects from dominating the final prediction.

B. Pixel-Domain Frequency-Aware Diffusion Purification

Adversarial examples are generated by adding weakly perceptible yet structured perturbations to the input. Owing to their strong denoising capability and stable generation behavior, diffusion models have been widely adopted for AP. In standard diffusion models, the forward process injects Gaussian noise whose spectrum is approximately uniform and weakly structured. In contrast, adversarial perturbations are typically more structured and tend to induce more salient responses in the mid-to-high-frequency range. Consequently, when adversarial inputs are processed by a diffusion-based purifier, the reverse denoising network must predict a more complex composite noise that consists of both the injected Gaussian noise and the adversarial perturbation.

To this end, a PFDP module is realized by implementing the reverse diffusion denoiser with an FD-UNet. FD-UNet performs adaptive frequency-domain filtering to expose perturbation-salient high-frequency responses and injects them into the decoder via skip connections as explicit auxiliary cues for adversarial-component prediction. The complementary features are propagated along the main pathway to estimate the residual stochastic noise and maintain semantic structures for stable reverse denoising.

As shown in Fig. 4, in FD-UNet, the input feature map is first processed by a ResNet block to obtain an intermediate representation $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$. The adaptive filter (AF) module then transforms $\mathbf{F}$ into the frequency domain

$$\mathbf{F}_{freq} = \mathcal{F}_r(\mathbf{F}) \tag{3}$$

where $\mathcal{F}_r(\cdot)$ denotes the 2-D real Fourier transform. Since $\mathbf{F}$ is real-valued, $\mathcal{F}_r(\cdot)$ returns only the one-sided nonredundant complex spectrum, while the omitted frequency components are implicitly determined by Hermitian symmetry.

Fixed frequency cutoffs are often too coarse to accommodate the diverse spectral profiles of different attacks. Therefore, each AF module maintains N learnable complex-valued frequency-domain templates $\{\mathbf{G}_i\}_{i=1}^{N}$. Each template consists of real and imaginary components and is matched to the resolution of the corresponding feature map. The

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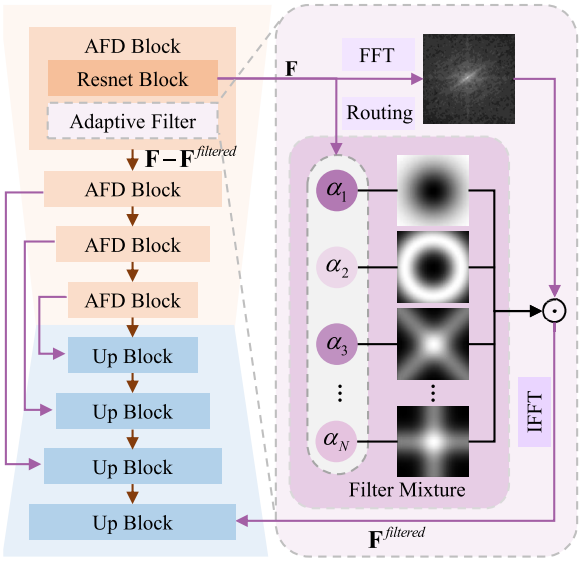


Fig. 4. Structure of the proposed FD-UNet with AFD blocks. Each AFD block consists of a ResNet block and an AF module, which performs frequency-domain filtering with learnable filter mixtures and dynamic routing to assist the prediction of adversarial-related components.

templates are initialized from a Gaussian distribution and are optimized end-to-end during training, without using hand-crafted frequency masks. To adapt the filtering behavior to the current input, a routing function generates samplewise and channelwise mixture weights over the learnable frequency-domain templates by applying global average pooling (GAP) followed by a lightweight multilayer perceptron (MLP)

$$\alpha = \mathrm{softmax}\big(\mathrm{MLP}\big(\mathrm{GAP}(\mathbf{F})\big)\big)$$
$$\alpha_{c,i} \in [0,1], \quad \sum_{i=1}^{N} \alpha_{c,i} = 1 \tag{4}$$

where $\alpha_{c,i}$ denotes the mixture weight assigned to template $\mathbf{G}_i$ for channel c.

Given the routing weights $\alpha_{c,i}$, the AF module forms a learnable filter mixture to adaptively emphasize perturbation-salient frequency responses. For each channel c, the frequency-domain feature $\mathbf{F}_{freq,c}$ is filtered by a weighted combination of templates $\mathbf{G}_i$, yielding the channelwise filtered spectrum

$$\mathbf{F}_{freq,c}^{filtered} = \sum_{i=1}^{N} \alpha_{c,i} \left(\mathbf{G}_i \odot \mathbf{F}_{freq,c}\right) \tag{5}$$

where $\odot$ is the elementwise multiplication. Aggregating all channels gives $\mathbf{F}_{freq}^{filtered}$, which is then mapped back to the spatial domain via inverse real Fourier transform

$$\mathbf{F}^{filtered} = \mathcal{F}_r^{-1}\left(\mathbf{F}_{freq}^{filtered}; H, W\right) \tag{6}$$

where $\mathcal{F}_r^{-1}(\cdot)$ denotes the inverse real Fourier transform with the original spatial size (H, W) specified. It reconstructs the full Hermitian spectrum from the one-sided representation and therefore produces a real-valued spatial feature map.

The filtered output $\mathbf{F}^{filtered}$ is designed to expose perturbation-salient high-frequency responses and is injected into the decoder via skip connections as explicit auxiliary cues, facilitating the identification and prediction of adversarial-related components in the composite noise. The complementary component is obtained by residual decomposition

$$\mathbf{F}^{latent} = \mathbf{F} - \mathbf{F}^{filtered}. \tag{7}$$

$\mathbf{F}^{latent}$ retains the remaining content, including most semantic structures together with the residual stochastic noise, and is propagated along the main pathway to support stable denoising and structure preservation.

The reverse diffusion training is further regularized with a multidomain consistency constraint across the pixel, structural, and frequency domains, realized by four complementary losses for reconstruction, structural similarity, spectral consistency, and high-frequency regularization.

To enforce pixel-level fidelity between the purification output and the clean sample, a mean squared error (mse) loss is adopted

$$\mathcal{L}_{\mathrm{mse}} = \frac{1}{N}\sum_{i=1}^{N} (\hat{x}_i - x_i)^2 \tag{8}$$

where $\hat{x}_i$ and x_i denote the ith element of the purified image and the corresponding clean image, respectively, and N is the total number of elements. This term provides a stable global constraint that limits large-scale deviations in the pixel space during denoising. However, using mse alone tends to average out fine textures, often resulting in overly smooth reconstructions and weakened detail.

To further preserve structural consistency, a multiscale structural similarity loss based on MS-SSIM [38] is adopted. MS-SSIM is computed between the purified image $\hat{x}$ and its corresponding clean target x. Both images are normalized to $[0, 1]$, and the structural loss is defined as follows:

$$\mathcal{L}_{\mathrm{struct}} = 1 - \mathrm{MS-SSIM}\big(\mathrm{norm}\,(\hat{x})\,, \mathrm{norm}\,(x)\big) \tag{9}$$

where norm$(\cdot)$ denotes min–max normalization to $[0, 1]$, and MS-SSIM measures multiscale structural consistency, with larger values indicating higher similarity. Minimizing 1 – MS-SSIM encourages the purification to retain structure-sensitive patterns, such as edges and textures, which are important for preserving RS semantic information.

Since adversarial perturbations often lead to larger deviations in the mid-to-high-frequency spectrum in the RS setting, a frequency-domain consistency loss is imposed on the amplitude spectra to suppress residual artifacts. The mid-to-high-frequency components are weighted through a continuous radial mask, where larger penalties are assigned to spectral components farther from the frequency center. The loss is defined as follows:

$$\mathcal{L}_{\mathrm{freq}} = \frac{1}{N}\left\| M_{\mathrm{HF}} \odot \left(\hat{A} - A\right)\right\|_2^2$$
$$M_{\mathrm{HF}}(u,v) = 1 + \sqrt{\left(\frac{u}{H} - \frac{1}{2}\right)^2 + \left(\frac{v}{W} - \frac{1}{2}\right)^2} \tag{10}$$

where $A = |\mathcal{F}(x)|$ and $\hat{A} = |\mathcal{F}(\hat{x})|$ denote the amplitude spectra of the clean image x and the reconstruction $\hat{x}$, respectively, and (u, v) are the frequency coordinates.

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To encourage the skip features to predominantly capture perturbation-salient high-frequency cues, a high-frequency energy ratio regularizer is imposed. Given the feature from the high-frequency branch $\{\mathbf{Z}_j\}$, the loss is defined as follows:

$$\mathcal{L}_{\text{hf}} = 1 - \frac{\sum_j \sum_{\omega \in \Omega_{\text{HF}}} \left| \mathcal{F}\left(\mathbf{Z}_j\right)(\omega) \right|^2}{\sum_j \sum_{\omega} \left| \mathcal{F}\left(\mathbf{Z}_j\right)(\omega) \right|^2} \tag{11}$$

where $\mathbf{Z}_j$ denotes the high-frequency features extracted from the frequency-aware skip branch of the jth adaptive frequency downsampling (AFD) block. Ω_{HF} denotes a predefined high-frequency region that serves as a coarse regularization prior. After shifting the low-frequency components to the spectrum center, the outer region farther from the center is regarded as the high-frequency region. By minimizing $\mathcal{L}_{\text{hf}}$, the fraction of spectral energy within Ω_{HF} is increased, thereby encouraging the skip-connected high-frequency branch to concentrate more on perturbation-salient high-frequency cues. It should be noted that Ω_{HF} is only used to regularize the energy distribution of the skip features, while the effective frequency bands emphasized in $\mathbf{F}^{filtered}$ are still adaptively selected by the learnable filter mixtures.

The overall objective is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda_{\text{struct}}\mathcal{L}_{\text{struct}} + \lambda_{\text{freq}}\mathcal{L}_{\text{freq}} + \lambda_{\text{hf}}\mathcal{L}_{\text{hf}}. \tag{12}$$

Specifically, $\mathcal{L}_{\text{mse}}$ enforces global pixel-level fidelity, $\mathcal{L}_{\text{struct}}$ encourages structural consistency, $\mathcal{L}_{\text{freq}}$ aligns amplitude spectra with higher penalties on mid-to-high frequencies, and $\mathcal{L}_{\text{hf}}$ regularizes skip-routed features to emphasize perturbation-salient high-frequency cues. Together, these complementary terms jointly regularize reverse diffusion, reducing residual artifacts and oversmoothing while preserving structurally and semantically meaningful content in RS imagery.

C. AVCD Strategy

Although PFDP substantially suppresses input-level adversarial corruption, residual effects may still persist in intermediate representations and become amplified through subsequent feature extraction. Such residual influence often concentrates in a subset of channels, leading to abnormal amplification or suppression of activations that can bias the downstream classifier even when the purified image remains visually plausible.

Accordingly, a feature-domain defense termed AVCD is introduced to further mitigate residual adversarial influence after PFDP. AVCD leverages two complementary signals. A feature-shift identification component quantifies how far the shallow representation deviates from a learned unshifted feature distribution, while a channel vulnerability estimate measures each channel's susceptibility to adversarial perturbations.

To instantiate the feature-shift identification component, shallow feature statistics are first modeled using an unshifted reference set. Specifically, the reference set is constructed by combining clean samples with lightly perturbed samples generated by injecting low-amplitude Gaussian noise, so that natural feature variations are covered while semantic labels remain unchanged

$$\mathcal{D}_{\text{unshifted}} = \mathcal{D}_{\text{clean}} \cup \mathcal{D}_{\text{noise}}. \tag{13}$$

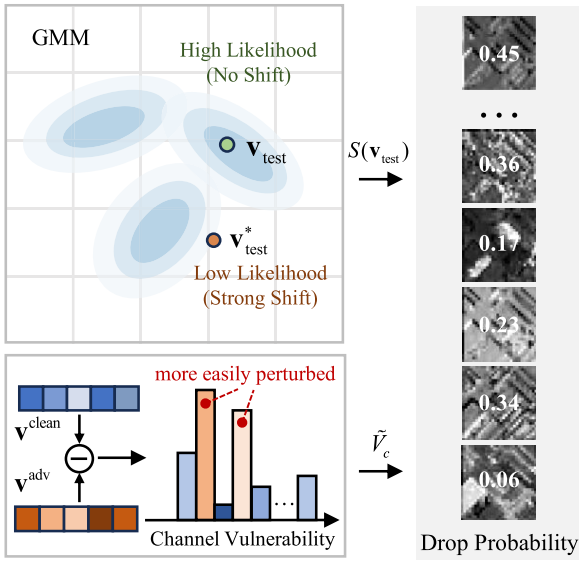


Fig. 5. Illustration of AVCD. The GMM-based likelihood quantifies the sample-level feature shift score $S(\mathbf{v}_{\text{test}})$, while channel vulnerability $\tilde{V}_c$ is estimated from clean–adversarial response differences. Both cues jointly determine channelwise dropout to suppress vulnerable channels.

For each sample in $\mathcal{D}_{\text{unshifted}}$, a shallow feature map $\mathbf{F}_{\text{unshifted}}$ is extracted from the victim network. A compact channel descriptor $\mathbf{v}$ is then formed by applying global max pooling over the spatial dimensions, which retains the strongest activation in each channel. Since the unshifted descriptors $\mathbf{v}$ are collected from multiple semantic categories, their distribution is typically multicluster rather than unimodal. Therefore, a GMM is adopted to characterize the unshifted reference distribution

$$p(\mathbf{v}) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(\mathbf{v} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right) \tag{14}$$

where K is the number of mixture components, π_k is the mixture weight with $\sum_{k=1}^{K} \pi_k = 1$, and $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ denote the mean and covariance of the kth component.

Given a test sample purified by PFDP, its shallow salient feature vector $\mathbf{v}_{\text{test}}$ is extracted and evaluated under the learned unshifted distribution. The corresponding log-likelihood is computed as follows:

$$LL(\mathbf{v}_{\text{test}}) = \log p(\mathbf{v}_{\text{test}}). \tag{15}$$

A higher likelihood indicates that $\mathbf{v}_{\text{test}}$ is well explained by the unshifted distribution, whereas a lower likelihood suggests a larger deviation. The feature shift score is then defined as the standardized negative log-likelihood

$$S\left(\mathbf{v}_{\text{test}}\right) = \text{Norm}\left(-LL\left(\mathbf{v}_{\text{test}}\right)\right) \tag{16}$$

where Norm$(\cdot)$ denotes z-score standardization. The mean and standard deviation are computed from the training set of GMM. These statistics are precomputed and kept fixed during testing.

As illustrated in Fig. 5, if a test feature $\mathbf{v}_{\text{test}}$ falls into a high-likelihood region, it is likely to be unshifted and yields a small shift score. Conversely, features located in low-density regions produce lower likelihood and thus a larger shift score

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$S(\mathbf{v}_{\text{test}})$. This score provides a sample-level measure of residual adversarial influence after PFDP and is used to modulate the strength of feature-domain suppression.

In addition, feature deviations are not necessarily uniform across channels. To identify the channels that are more sensitive to adversarial perturbations, paired shallow features are collected from clean and adversarial inputs, yielding pooled descriptors $\mathbf{v}^{\text{clean}}$ and $\mathbf{v}^{\text{adv}}$. The perturbation-induced activation change of channel c is defined as follows:

$$\Delta_c = \left|\mathbf{v}_c^{\text{clean}} - \mathbf{v}_c^{\text{adv}}\right|. \tag{17}$$

The channel vulnerability score is estimated by averaging the response change over paired clean–adversarial samples

$$\begin{aligned} V_c &= \mathbb{E}\left[\Delta_c\right] \\ \tilde{V}_c &= \frac{V_c}{\max_j\left(V_j\right)} \end{aligned} \tag{18}$$

where V_c measures how sensitive channel c is to adversarial perturbations, and a larger value indicates a more easily perturbed channel. The normalized score $\tilde{V}_c$ serves as a vulnerability weight across channels.

AVCD determines channel suppression by combining the samplewise shift score $S(\mathbf{v}_{\text{test}})$ and the channel vulnerability $\tilde{V}_c$. Specifically, the dropout probability for channel c is defined as follows:

$$P_{drop}\left(c \mid \mathbf{v}_{\text{test}}\right) = max\left(\sigma\left(S\left(\mathbf{v}_{\text{test}}\right) \cdot \tilde{V}_c\right) - 0.5, 0\right) \tag{19}$$

where $\sigma(\cdot)$ is the sigmoid function. This formulation assigns higher dropout probabilities to channels that are both more vulnerable and associated with a more shifted sample representation. The channelwise dropout operation is then performed as follows:

$$\widetilde{\mathbf{F}}_c = \begin{cases} 0, & \text{with probability } P_{\text{drop}}\left(c \mid \mathbf{v}_{\text{test}}\right) \\ \mathbf{F}_c, & \text{otherwise} \end{cases} \tag{20}$$

where $\mathbf{F}_c$ and $\widetilde{\mathbf{F}}c$ denote the original and suppressed features of channel c, respectively. As illustrated in Fig. 5, $S(\mathbf{v}_{\text{test}})$ and $\tilde{V}_c$ jointly control P_{drop}, enabling AVCD to selectively suppress adversarially vulnerable channels at inference time.

AVCD complements PFDP by refining intermediate representations using channelwise dropout guided by feature shift and vulnerability, providing an additional layer of protection for RS scene classification under adversarial conditions.

IV. Experiments

A. Experimental Setup

1) Dataset Description: Experiments are conducted on two widely used benchmark RS scene classification datasets, namely, UCM land-use [20] and AID [21]. UCM contains 2100 RGB aerial scene images from 21 land-use categories, with 100 images per class. All images are of size 256 × 256 pixels and have a spatial resolution of approximately 0.3 m/pixel. AID is a larger and more challenging benchmark, consisting of 10 000 aerial images spanning 30 scene categories collected from Google Earth. Each category contains approximately 220–420 images, and all samples are provided at a fixed image size of 600 × 600 pixels. Compared with UCM, AID exhibits substantially greater variation in spatial resolution, ranging from 0.5 to 8 m/pixel, together with diverse imaging conditions, which makes robust scene classification more demanding.

2) Attack Methods: To evaluate robustness under different attack settings, adversarial examples are generated using five widely adopted methods, including FGSM [22], I-FGSM [23], PGD [14], CW [24], and Jitter [25]. For gradient-sign attacks, the perturbation budget is fixed to $\epsilon = 8/255$. I-FGSM and PGD are iterative attacks with a step size $\alpha = 2/255$ and ten iterations. PGD additionally uses random initialization to strengthen the attack. The CW attack is implemented as an optimization-based method with $c = 1$, $\kappa = 0$, 50 optimization steps, and a learning rate of 0.01. Jitter is evaluated with $\epsilon = 8/255$, $\alpha = 2/255$, and ten iterations, and it also uses random initialization. These configurations cover fast single-step perturbations, stronger iterative attacks, and optimization-based attacks, enabling a comprehensive robustness evaluation. To further evaluate robustness under remote-sensing-specific adversarial scenarios, Mixup-Attack [39] and Mixcut-Attack [39] are additionally introduced as supplementary attacks.

3) Victim Models and Comparison Methods: Robustness is evaluated on three representative victim networks that cover different architectural styles and capacity levels, including ResNet-50, MobileNet-V3, and DenseNet-121. Using multiple backbones helps verify that the defense performance is not tied to a specific model type. For adversarial defense comparison, the proposed DDAP is compared with four competitive defense methods, including APMH [16], AToP [17], UAD-RS [18], and OSCP [19]. All these methods are tested under the same attack configurations and evaluation metrics to ensure a fair and consistent comparison.

4) Evaluation Metrics: Defense performance is evaluated on both clean and adversarial inputs. Standard accuracy is defined as the accuracy on clean images, while robust accuracy refers to that on adversarial images. In both cases, accuracy is computed as the number of correctly classified samples divided by the total number of test samples. As shown in Table I, the victim models suffer a pronounced drop in accuracy under the attacks. For instance, on UCM, ResNet-50 drops from 96.67% on clean images to 2.86% under I-FGSM and PGD, similar trends are observed on AID, demonstrating the severity of adversarial perturbation and the need for effective defenses.

5) Implementation Details: All RS images are resized to 224 × 224 before being fed into the models. For each dataset, 50% of the samples are randomly selected as the training set, and the remaining 50% are used as the test set. During training, the PFDP optimization objective is defined as a weighted combination of four loss terms, namely, $\mathcal{L}_{\text{mse}}$, $\mathcal{L}_{\text{struct}}$, $\mathcal{L}_{\text{freq}}$, and $\mathcal{L}_{\text{hf}}$, with the weights set to 1.0, 0.6, 0.4, and 0.1, respectively. The PFDP module is trained for 200 epochs using the AdamW optimizer with an initial learning rate of 1×10^{-4}, and the batch size is set to 8. All experiments are implemented using the PyTorch deep learning framework and conducted on NVIDIA GeForce RTX 3090 GPUs.

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TABLE I
BASELINE ACCURACY (%) OF VICTIM MODELS UNDER DIFFERENT ATTACKS WITHOUT DEFENSE

Dataset	Victim Models	Standard acc.	Robust acc.				
		Clean	FGSM	I-FGSM	CW	PGD	Jitter
UCM	MobileNet-V3	94.29	18.67	0.00	0.00	0.00	0.29
	ResNet-50	96.67	65.71	2.86	11.14	2.86	16.19
	DenseNet-121	97.05	50.10	0.67	4.48	0.76	8.38
AID	MobileNet-V3	92.44	12.20	0.02	0.00	0.00	0.26
	ResNet-50	94.78	39.08	0.00	0.50	0.00	7.80
	DenseNet-121	95.26	38.86	0.02	0.18	0.02	6.04

TABLE II
ROBUST ACCURACY (%) OF DIFFERENT METHODS AGAINST VARIOUS ATTACKS ON THE UCM AND AID DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Victim Models	Defense	Attack on UCM dataset					Attack on AID dataset				
		FGSM	I-FGSM	PGD	CW	Jitter	FGSM	I-FGSM	PGD	CW	Jitter
ResNet-50	APMH [16]	77.71	78.00	79.05	86.10	86.48	50.10	42.20	45.32	69.50	56.12
	AToP [17]	82.10	83.81	85.24	90.19	88.57	61.00	63.86	66.70	82.24	74.88
	UAD-RS [18]	83.71	86.57	86.95	90.38	88.86	70.76	74.16	80.86	87.32	82.56
	OSCP [19]	84.38	86.76	81.24	90.67	88.38	64.08	65.28	63.46	69.92	69.16
	DDAP	**85.14**	**90.00**	**90.19**	**91.62**	**91.24**	**74.20**	**78.74**	**81.66**	**88.48**	**84.90**
MobileNet-V3	APMH [16]	34.95	32.86	43.24	64.86	40.48	20.92	22.00	21.40	58.94	32.32
	AToP [17]	47.24	56.67	61.33	75.81	67.14	34.38	47.20	50.08	71.94	62.14
	UAD-RS [18]	38.95	57.52	66.10	83.81	75.81	24.48	36.04	46.02	80.38	68.34
	OSCP [19]	44.86	47.62	58.86	72.00	59.24	**39.74**	39.96	51.08	50.94	54.12
	DDAP	**53.14**	**65.33**	**70.19**	**84.48**	**79.05**	39.16	**50.16**	**58.62**	**81.90**	**73.68**
DenseNet-121	APMH [16]	68.67	71.52	72.48	87.33	84.57	45.72	25.56	31.06	67.48	50.92
	AToP [17]	75.24	78.57	80.38	88.76	84.19	58.92	61.70	63.28	81.16	72.58
	UAD-RS [18]	76.29	82.48	85.62	91.05	89.81	60.08	66.36	72.68	87.34	80.76
	OSCP [19]	79.91	83.81	83.71	90.57	84.67	63.94	68.40	59.38	73.00	61.52
	DDAP	**81.71**	**85.33**	**87.43**	**92.67**	**91.05**	**65.86**	**70.66**	**76.00**	**87.66**	**82.80**

B. Robustness Comparison

Across both UCM and AID, the proposed DDAP consistently achieves the best or competitive robust scene classification accuracy under most evaluated attacks and victim models. As shown in Table II, on UCM, DDAP delivers stable gains over comparative defense methods under iterative and stronger attacks, such as I-FGSM and PGD. For example, with ResNet-50, DDAP reaches 90.00% and 90.19% robust accuracy under I-FGSM and PGD, respectively, outperforming the strongest methods by a clear margin. Similar improvements are observed on DenseNet-121, where DDAP attains 87.43% accuracy under PGD and 92.67% accuracy under CW. In particular, DDAP remains effective on the lightweight MobileNet-V3, where robustness is typically harder to maintain. Under PGD and I-FGSM on UCM, DDAP achieves 70.19% and 65.33% accuracy, showing a substantial advantage over prior methods and suggesting suitability for resource-constrained deployment.

On the more challenging AID benchmark, all defenses exhibit lower accuracy due to larger intraclass variation and cross-source, multiresolution imaging conditions, yet DDAP remains the most competitive and stable method across attacks. With ResNet-50, DDAP achieves 78.74% and 81.66% under I-FGSM and PGD, respectively, and maintains strong performance under CW and Jitter. The advantage is also clear on the MobileNet-V3 and DenseNet-121 models, where DDAP attains 58.62% and 76.00% under PGD, surpassing all compared methods. These results indicate that DDAP generalizes well across datasets, architectures, and heterogeneous attack types.

C. Qualitative Purification Results

Figs. 6 and 7 present qualitative purification results on the UCM and AID datasets under PGD attacks, enabling direct visual comparison between the purification performance of DDAP and other defense methods. The examples are chosen to emphasize texture-rich regions and sharp boundaries, where adversarial perturbations tend to be entangled with fine scene details.

APMH and AToP can reduce visible perturbation patterns, but they tend to leave residual artifacts or introduce noticeable blur and local contrast inconsistencies in the enlarged regions. UAD-RS generally yields cleaner reconstructions, yet mild blurring can still be observed on thin structures, such as

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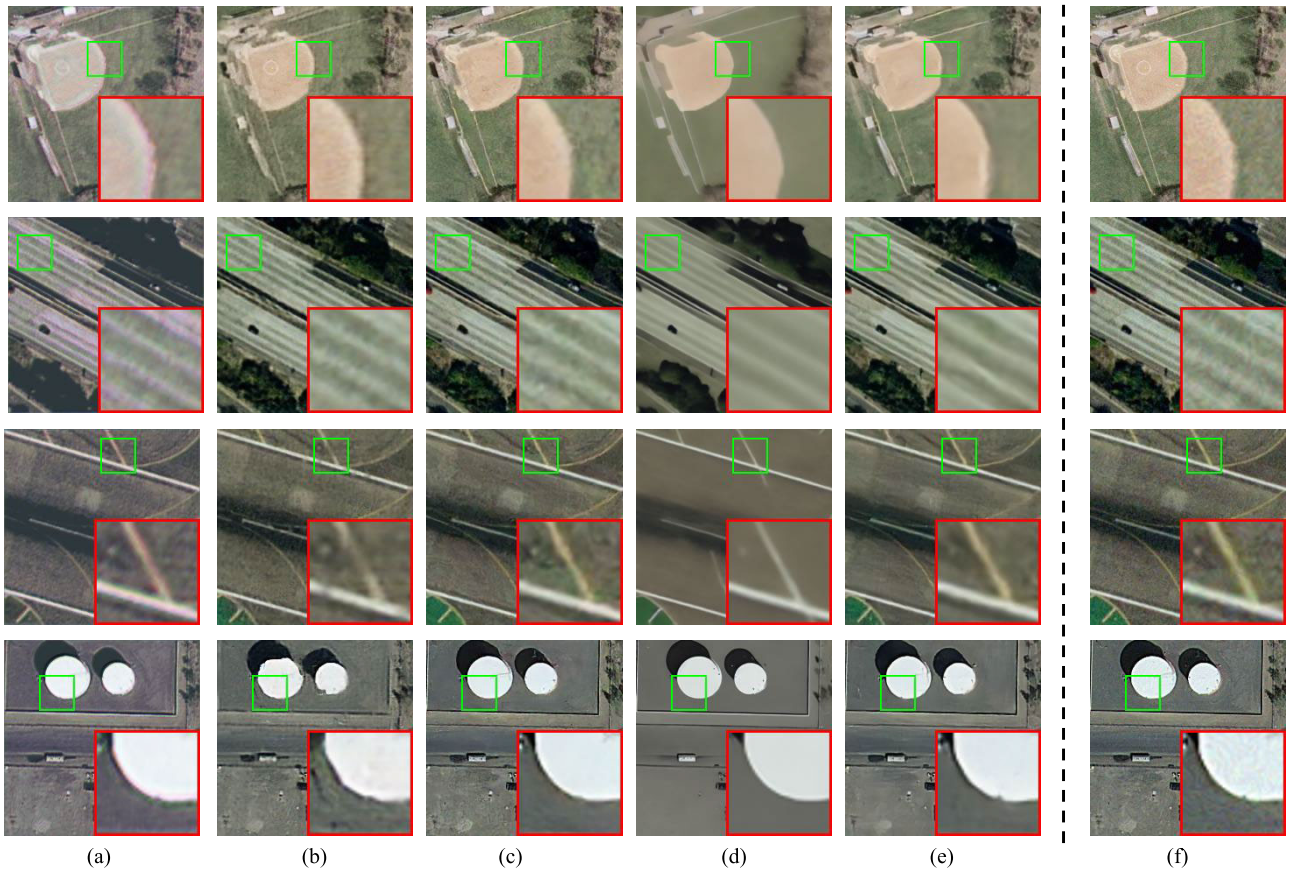


Fig. 6. Pixel-domain purification results on the UCM dataset. The adversarial examples are generated using PGD attack on the ResNet-50 victim model. (a)–(e) Purified outputs of APMH, AToP, UAD-RS, OSCP, and DDAP, while (f) corresponding adversarial input. Local regions are highlighted and enlarged to facilitate visual comparison, emphasizing the differences in perturbation suppression and detail preservation across methods.

lane markings, building edges, and small rooftop patterns. OSCP shows the strongest smoothing effect, which suppresses perturbations but also removes discriminative details, leading to flat textures and weakened boundaries, especially on the more complex AID scenes. In contrast, DDAP produces pixel-domain reconstructions that better preserve geometric boundaries and texture fidelity while effectively suppressing adversarial artifacts, as evidenced by clearer line structures and sharper object edges in the zoomed regions.

D. Clean–Robust Balance

In addition to accuracy under attack, the impact on clean inputs is examined to assess whether a defense overcorrects and harms normal recognition. As reported in Table III, on the UCM dataset with the ResNet-50 victim model, DDAP preserves a high standard accuracy across all attack settings, remaining above 91.00% with only a 3.53%–5.53% drop from the no-defense baseline of 96.67%, while simultaneously delivering large gains on adversarial inputs. For instance, under I-FGSM, DDAP achieves 93.14% standard accuracy and 90.00% robust accuracy, and under PGD, it achieves 92.57% standard accuracy and 90.19% robust accuracy. Without defense, robust accuracy drops to 2.86% under both I-FGSM and PGD. By contrast, OSCP incurs a substantially larger clean–accuracy penalty, such as a 14.29% drop under PGD, indicating that overly aggressive smoothing can remove discriminative scene details together with perturbations. These results suggest that DDAP offers a favorable clean–robust tradeoff. This is consistent with its design, where PFDP suppresses perturbations while maintaining semantic structures, and AVCD is applied selectively, so that stable semantic features are less affected on clean samples.

E. Remote-Sensing-Specific Attack Evaluation

To further evaluate the generalization ability of the proposed DDAP under remote-sensing-specific adversarial scenarios, two adversarial attacks designed for RS, namely, Mixup-Attack [39] and its variant Mixcut-Attack [39], are introduced. In the experiments, both attacks are conducted in a black-box transfer setting, where adversarial examples are generated using ResNet-18 as the surrogate model and then transferred to the ResNet-50 victim model. For all defense methods, the corresponding purification models are trained using PGD-based adversarial examples.

The results are reported in Table IV. On the UCM dataset, DDAP achieves robust accuracies of 89.71% and 90.00% under Mixup-Attack and Mixcut-Attack, respectively, outperforming the compared methods. On the more challenging AID dataset, DDAP also shows competitive robustness, reaching 85.24% under Mixup-Attack and 85.58% under

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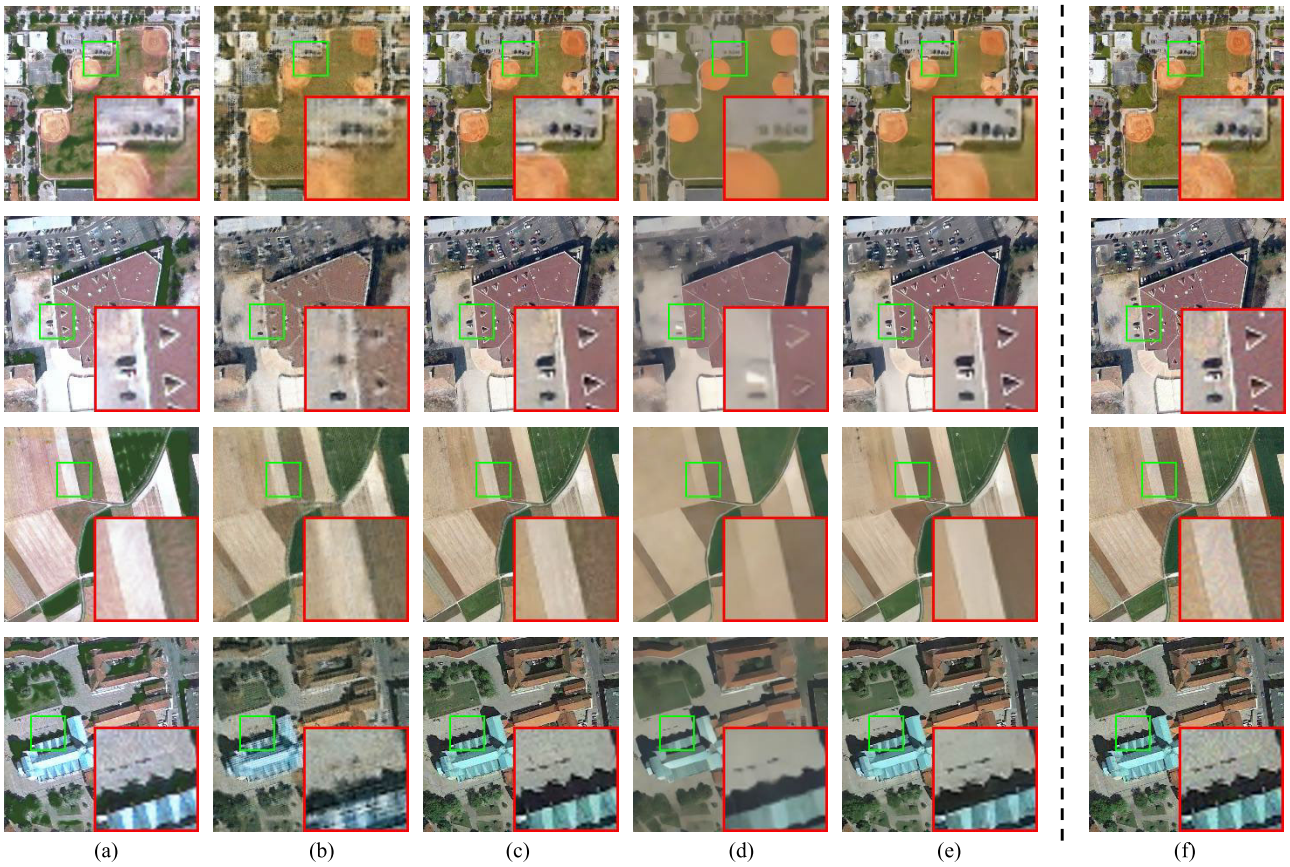


Fig. 7. Pixel-domain purification results on the AID dataset. The adversarial examples are generated using PGD attack on the DenseNet-121 victim model. (a) APMH. (b) AToP. (c) UAD-RS. (d) OSCP. (e) DDAP. (f) Adv.

Mixcut-Attack. These results further demonstrate that DDAP generalizes well not only to conventional adversarial attacks but also to remote-sensing-specific attack methods.

F. Ablation Experiments

Under the I-FGSM attack with the UCM dataset, the ablation results verify the complementarity between PFDP and AVCD, as shown in Table V. When no defense is applied, the three victim models almost completely fail, with the accuracies of 2.86%, 0.00%, and 0.67% for ResNet-50, MobileNet-V3, and DenseNet-121, respectively, indicating the strong disruptive effect of I-FGSM. Enabling AVCD alone yields recovery to 50.95%, 15.81%, and 54.29%, suggesting that feature-domain channel suppression can partially block the influence of a set of abnormal activations, but its effectiveness remains limited. Meanwhile, PFDP alone restores performance substantially to 87.71%, 57.33%, and 83.05%, demonstrating that frequency-aware diffusion purification removes the dominant corruption and recovers discriminative structures. Adding AVCD on top of PFDP further improves the accuracies to 90.00%, 65.33%, and 85.33%. PFDP provides the main robustness recovery in the pixel domain, while AVCD further strengthens robustness by suppressing residual adversarial effects in the feature space without network training.

As shown in Fig. 8, to further analyze the frequency-domain behavior of PFDP, the radial power spectrum plot provides

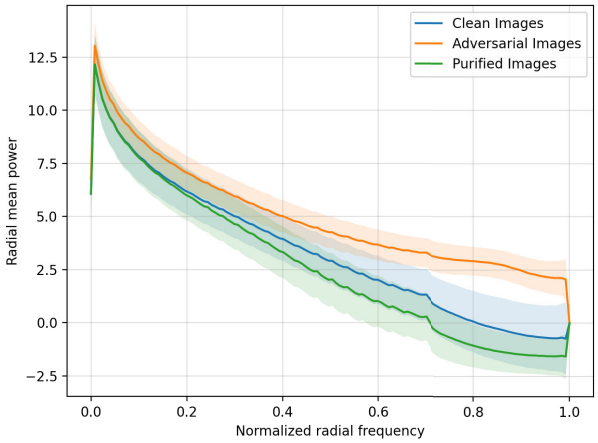


Fig. 8. Radial power spectrum curves.

a frequency-domain view of how adversarial perturbations and purification affect the overall energy distribution. On the UCM dataset under I-FGSM attack against ResNet-50, the adversarial curve consistently lies above the clean curve over a broad mid-to-high frequency range, indicating that the attack injects additional structured energy that is especially pronounced at higher frequencies. After purification of PFDP, the spectrum is pulled back toward the clean curve, with the largest correction occurring in the mid-to-high-frequency region where adversarial effects are most prominent. This behavior

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TABLE III

STANDARD AND ROBUST ACCURACY (%), WITH PARENTHESES REPORTING CHANGES RELATIVE TO "NO DEFENSE"

Attack Method	Defense	Standard acc.	Robust acc.
FGSM	No defense	96.67	65.71
	APMH [16]	90.48 (-6.19)	77.71 (+12.00)
	AToP [17]	**92.48 (-4.19)**	82.10 (+16.39)
	UAD-RS [18]	90.95 (-5.72)	83.71 (+18.00)
	OSCP [19]	85.33 (-11.34)	84.38 (+18.67)
	DDAP	91.14 (-5.53)	**85.14 (+19.43)**
I-FGSM	No defense	96.67	2.86
	APMH [16]	86.57 (-10.10)	78.00 (+75.14)
	AToP [17]	92.48 (-4.19)	83.81 (+80.95)
	UAD-RS [18]	90.95 (-5.72)	86.57 (+83.71)
	OSCP [19]	87.05 (-9.62)	86.76 (+83.90)
	DDAP	**93.14 (-3.53)**	**90.00 (+87.14)**
PGD	No defense	96.67	2.86
	APMH [16]	91.05 (-5.62)	79.05 (+76.19)
	AToP [17]	92.48 (-4.19)	85.24 (+82.38)
	UAD-RS [18]	90.95 (-5.72)	86.95 (+84.09)
	OSCP [19]	82.38 (-14.29)	81.24 (+78.38)
	DDAP	**92.57 (-4.10)**	**90.19 (+87.33)**
CW	No defense	96.67	11.14
	APMH [16]	91.52 (-5.15)	86.10 (+74.96)
	AToP [17]	**92.48 (-4.19)**	90.19 (+79.05)
	UAD-RS [18]	90.95 (-5.72)	90.38 (+79.24)
	OSCP [19]	91.14 (-5.53)	90.67 (+79.53)
	DDAP	92.00 (-4.67)	**91.62 (+80.48)**
Jitter	No defense	96.67	16.19
	APMH [16]	92.38 (-4.29)	86.48 (+70.29)
	AToP [17]	92.48 (-4.19)	88.57 (+72.38)
	UAD-RS [18]	90.95 (-5.72)	88.86 (+72.67)
	OSCP [19]	86.38 (-10.29)	88.38 (+72.19)
	DDAP	**93.05 (-3.62)**	**91.24 (+75.05)**

TABLE IV

ROBUST ACCURACY (%) OF DIFFERENT DEFENSE METHODS UNDER REMOTE-SENSING-SPECIFIC ATTACKS

Datasets	Attacks	APMH	AToP	UAD-RS	OSCP	DDAP
UCM	Mixup	82.95	83.33	86.86	79.81	**89.71**
	Mixcut	82.29	83.81	86.19	80.96	**90.00**
AID	Mixup	44.76	65.50	83.16	62.78	**85.24**
	Mixcut	45.14	65.44	83.44	63.06	**85.58**

TABLE V

ABLATION STUDY OF THE KEY MODULES IN THE DDAP

Attack	PFDP	AVCD	ResNet-50	MobileNet-V3	DenseNet-121
✓			2.86	0.00	0.67
✓		✓	50.95	15.81	54.29
✓	✓		87.71	57.33	83.05
✓	✓	✓	**90.00**	**65.33**	**85.33**

suggests that the purification process effectively suppresses perturbation-induced high-frequency components rather than only improving pixel-domain similarity. Meanwhile, the three curves remain relatively close in the low-frequency region, implying that global intensity trends and coarse scene layout are largely preserved. Notably, the purified spectrum becomes slightly lower than the clean spectrum near the highest frequencies, which is consistent with mild texture smoothing caused by denoising. The radial spectrum comparison supports the intended frequency-aware behavior of PFDP by reducing adversarially amplified high-frequency energy while preserving the dominant low-frequency content.

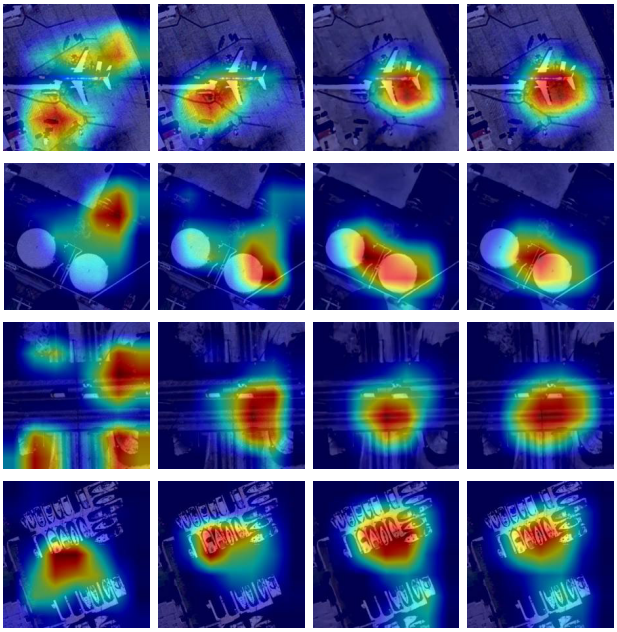


Fig. 9. Grad-CAM visualizations under different inputs and defenses. (a) Adv: adversarial image without defense. (b) Adv + AVCD: adversarial image with feature-domain AVCD only. (c) DDAP: adversarial image purified by the full DDAP pipeline, including pixel-domain PFDP followed by AVCD. (d) Clean: clean image as a reference.

To visually demonstrate the effectiveness of AVCD and the overall DDAP framework, Grad-CAM visualizations are employed to reveal how adversarial perturbations and defenses reshape the classifier's spatial attention patterns. Fig. 9 reports the Grad-CAM results on the UCM dataset under the I-FGSM attack with the ResNet-50 victim model, comparing "Adv," "Adv + AVCD," "DDAP," and "Clean." For adversarial inputs, attention maps tend to drift away from the semantically relevant regions and become spatially scattered, indicating that the perturbation has distorted the internal evidence used for prediction. After applying AVCD, the attention maps become noticeably more structured, suggesting that suppressing adversarially vulnerable channels can partially reduce abnormal feature amplification. However, the highlighted regions may still deviate from the clean attention pattern. When PFDP is combined with AVCD in DDAP, the resulting Grad-CAM responses are consistently closer to those of clean images, with stronger focus on object-related regions and better alignment with scene geometry. This behavior indicates that PFDP helps remove perturbation-induced interference, while the subsequent feature-domain channel suppression with AVCD

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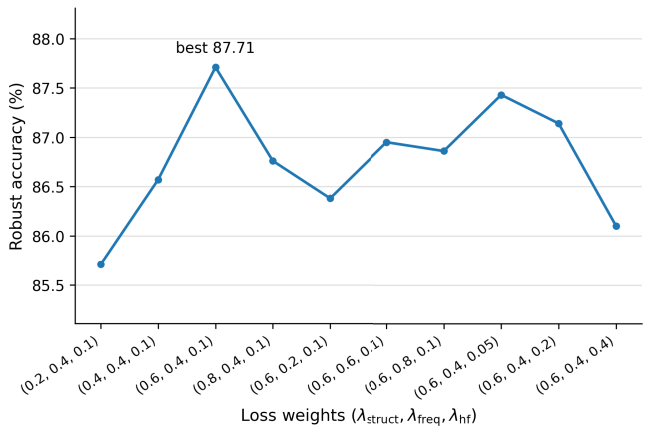


Fig. 10. Sensitivity of PFDP to the loss weights ($\lambda_{\text{struct}}, \lambda_{\text{freq}}, \lambda_{\text{hf}}$), with the y-axis showing the resulting robust accuracy.

TABLE VI
INFERENCE SPEED (FPS) AND MEAN ROBUST ACCURACY (%)

	APMH	AToP	UAD-RS	OSCP	DDAP
FPS	31.42	**46.47**	0.98	1.63	41.62
Mean robust acc. (%)	81.47	85.98	87.29	86.29	**89.64**

prevents remaining adversarial influence from dominating the final decision.

G. Parameter Analysis

To examine how the multidomain consistency terms in (12) affect PFDP and to select a stable setting, Fig. 10 presents a sensitivity study on the UCM dataset under the I-FGSM attack with the ResNet-50 victim model. The x-axis enumerates the weight triplets ($\lambda_{\text{struct}}, \lambda_{\text{freq}}, \lambda_{\text{hf}}$), and the y-axis reports the resulting robust accuracy.

When $\lambda_{\text{freq}} = 0.4$ and $\lambda_{\text{hf}} = 0.1$ are fixed, increasing λ_{struct} up to a moderate level steadily improves robustness, suggesting that an appropriate structural constraint helps preserve semantically meaningful edges and textures that support stable prediction. Pushing λ_{struct} further leads to a mild degradation, indicating that overly emphasizing perceptual structure may start to weaken the suppression of adversarial artifacts. With $\lambda_{\text{struct}} = 0.6$ fixed, varying λ_{freq} results in relatively stable performance, implying that spectral alignment is beneficial but should remain complementary rather than dominant. Adjusting λ_{hf} shows that a mild high-frequency feature regularization is helpful, whereas an overly large weight degrades robustness, likely because excessive emphasis on high-frequency energy can over-constrain skip features and suppress useful scene details. Overall, PFDP remains stable within a reasonable weight range, and the best performance of 87.71% is achieved at (0.6, 0.4, 0.1), which is adopted as the default setting.

H. Time Complexity Analysis

The time complexity under the unified evaluation setting is reported in Table VI. Purification speed in frames per second (fps) and mean robust accuracy averaged over five attacks are reported for five defenses on a ResNet-50 backbone. DDAP achieves the second-highest throughput at 41.62 fps while obtaining the best mean robust accuracy of 89.64%, indicating a favorable speed–robustness tradeoff. AToP attains the highest fps because its test-time procedure is close to a pure feed-forward pipeline without multistep iterative optimization, but its average robustness remains below DDAP. APMH is also relatively fast since, although it involves iterative updates, its computation is still lighter than traditional diffusion-based methods. In contrast, UAD-RS is the slowest because it relies on a multistep reverse denoising chain, which substantially increases inference cost. OSCP improves over multistep diffusion by reducing purification to a single inference step, yet it remains comparatively slow due to the use of large generative backbones and additional conditioning modules.

V. CONCLUSION

This article proposes a DDAP framework to improve the robustness of RS scene classification under adversarial attacks. DDAP mitigates adversarial effects within a unified pipeline at both the pixel and feature levels. In the pixel domain, PFDP is realized by implementing the reverse diffusion denoiser with FD-UNet, which explicitly models perturbation-related patterns and produces semantically faithful RS images under multidomain consistency constraints. In the feature domain, an AVCD strategy is introduced to suppress residual adversarial influence by combining a likelihood-based feature shift score with a channel vulnerability estimate, thereby preventing a subset of abnormal activations from dominating the final decision. Extensive experiments on UCM and AID across multiple victim backbones and attack types demonstrate that DDAP consistently improves robustness while maintaining a favorable clean–robust balance compared with representative defense baselines.

Despite these achievements, there are still some limitations that need to be addressed in future work. First, the current DDAP framework is mainly evaluated on RS scene classification, while its effectiveness on more complex dense prediction tasks, such as object detection and semantic segmentation, remains to be further investigated. Therefore, the dual-domain defense design will be extended to broader RS interpretation tasks in future work. Second, the current PFDP module mainly relies on generic consistency constraints for purification, which may not be fully aligned with the decision requirements of downstream tasks. To address this issue, task-aware purification will be explored by reformulating the consistency constraints toward task-guided objectives, so that the purification process can better support downstream prediction reliability.

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